Fusion Model for Image Classification and Enhancement

Goal: Build a better disease detection model by **fusing** (combining) different information.

For Examples:

- Leaf images + Fruit images
- RGB images + Thermal images
- CNN features + Vision Transformer features
- Multiple fruit types (tomato + potato + pepper …)

Phase 0 – Preparation (Before You Start Coding)

□ Understand what kind of fusion you want to try

- Early fusion → combine pictures/data at the beginning
- Late fusion → combine final predictions at the end
- Feature fusion → combine middle features from two models

□ Choose dataset folder structure Examples:

```
├── train
│   ├── Tomato_healthy
│   ├── Tomato_Bacterial_spot
│   └── …
├── val
└── test
```

or (for leaf + fruit fusion)

```
├── train_leaf
├── train_fruit
├── val_leaf
├── val_fruit etc.
```

□ Decide number of classes: _____ (e.g. 5, 10, 38)

□ Decide main backbone models you will compare Example choices:

- ResNet50
- EfficientNet
- ViT (Vision Transformer) small/base
- MobileNetV3 (fast)

Phase 1 – Data Preparation Checklist

□ Images resized to 224×224 (most models like this size)
□ Images normalized (ImageNet mean & std used)
□ Data augmentation ON for training only Common: rotate ±15°, flip horizontal, brightness/contrast change
□ Class imbalance checked → decided whether to use:

- Weighted sampler

- Class weights in loss
- Just ignore for first experiment

□ Train / Val / Test folders ready
□ Number of images per split written down: Train: _____ Val: _____ Test: _____

Phase 2 – Choose & Prepare ONE Experiment

Experiment A – Early fusion (concatenate images or channels)
Experiment B – Feature-level fusion (concat CNN + ViT features)
Experiment C – Late fusion (average / weighted average of two models)
Experiment D – Late fusion with stacking (use Logistic Regression on predictions) or Others
Chosen experiment: _____

Phase 3 – Model Building Steps

Step 3.1 – Load pre-trained model(s)
Step 3.2 – Remove / change the last classification layer
Step 3.3 – For early fusion → change first conv layer if needed (different input channels)
Step 3.4 – For feature fusion → connect two backbones → one new linear layer
Step 3.5 – For late fusion → prepare **two separate models**
Step 3.6 – Decide loss function → usually CrossEntropyLoss
Step 3.7 – Choose optimizer → usually Adam or AdamW
Step 3.8 – Choose learning rate → start with 1e-4 or 3e-4

Phase 4 – Training Workflow (Repeat for Each Model)

1. Put model on GPU (if available)
2. For each epoch:
   □ Train one full pass on train set
   □ Calculate train loss
   □ Evaluate on validation set → val loss + accuracy or F1
   □ Save model if validation result is best so far
3. After all epochs → keep the best model (lowest val loss or highest F1)

Phase 5 – Fusion-specific Actions

Early / Feature fusion
□ Train only **one model** → directly use it on test

Late fusion (simple average or weighted)
□ Train model A (e.g. leaf model) → get best weights
□ Train model B (e.g. fruit model) → get best weights
□ On validation set: try different weights (0.5+0.5, 0.6+0.4, 0.7+0.3 …) → choose best combination
□ Use that weight on test set

Late fusion with stacking
□ Get predictions (probabilities) of model A and B on validation set
□ Train small Logistic Regression or small neural net on these probabilities
□ Use it on test set

Phase 6 – Final Evaluation (Must Do)

□ Run best model(s) on **test set only once**

□ Calculate:

Metric	**Value**
Accuracy	____ %
Weighted F1	____
Macro F1	____
Per-class F1	list them

□ Compare with single-model baseline (very important!)

This is a sample format for Fusion Model Development. If you have any ideas or suggestions for improving the process, please feel free to share making any changes to the approach. Good Luck.